Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Numericals Energy storage devices

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Numericals Energy storage devices

1 Numericals Energy storage devices

Study focus: Numericals_Energy storage devices. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000050 · EV0000051

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices and we'll try to solve some numericals today on this topic.

Key details

- After that, you also need to find out what is the weight of the entire battery.
- So here first we have to find out the weight of zinc.
- And then we have to find out the weight of the entire battery.
- I into T, that is current into time.
- And two hours is the time.
- It is working for it is lasting for two hours.
- That is we are converting it into seconds.
- We need to be very careful with the units.
- In this case it is zinc.
- That is the anodic material.
- N is the number of electrons getting exchanged.
- We know from the reaction it is two electrons.
- So therefore C is equal to Wnf by M.
- From here we can find out the weight of the electroarms.
- material by rearranging this equation.
- That is the reason why we found out the capacity.
- In this case, we could determine capacity from here, and this formula is used to determine weight of the electroactive material.
- So, therefore, capacity is known.

- into 652 multiplied by divided by two multiplied by nine six five zero and we have found out the weight of zinc it is three point zero three grams of zinc.
- Now coming to the next part where we want to find out the weight of the entire battery.
- Amount of charge that can be stored in the battery per unit weight of the battery.
- And then there we said weight of the battery, when we say per unit weight of the battery, that is weight of the entire battery, including all the components.
- That is electricity storage density because we said the amount of charge that can be stored in the battery.
- And what is the amount of charge that can be stored in the battery?
- That is equal to the capacity of the battery.
- Only thing you need to be very, very careful with the units.
- The unit is where you have to be very careful.
- You have to give your answer in ampere hour per kg.
- This is the requirement of this numerical atomic mass of lithium is seven.

Example or application from the lecture

- Let us take up the first numerical on battery characteristics.
- And for this electricity storage density is given to us and molar mass of zinc is known, 65 grams, and F, that is Faraday's constant, is to be taken as 96500 Coulombs per mole.
- Let us see the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 05:00–09:59

Capacity is either current into time or it can be determined by WNF by M.

Key details

- So because no current or time is mentioned here, we have to go with the other formula that is C is equal to Wnf by the weight of the battery.
- It is 2 grams.
- Lithium is always we know lithium is monovalent so plus one electron therefore here we have one electron that is being exchanged in this and 9650 divided by seven that is the atomic mass of lithium.
- But this is in Coulombs.
- Remember capacity is always what when we calculate it will be in Coulombs and Coulombs is always ampere second.
- So therefore if you want to convert it into ampere hours what do you need to do?
- Divide it by three six zero and that would give you an answer seven point six five eight ampere hours is the capacity.
- Weight of the battery.
- Please read carefully.
- This is the weight of the electroactive material which is used to calculate capacity of the battery, and this weight is weight of the entire battery.

- It is the total weight of the battery and it contains 5 kg of lead as the anode material.
- for 10 hours.
- The time is known to us.
- Number of electron transferred in the redox reaction is 2.
- These are simple.
- elements for which may you should know what is the simple reaction when it undergoes oxidation.
- Since it's the unknown material, we write the reaction this way.
- We need the energy.
- Energy density and we need the power density.
- What do we have?
- So that means I can straight away determine the capacity here using C is equal to Wnf by M.
- Yes, so we will use C is equal to Wnf by M that is 5 into 10 raised to the power of 3.
- Why did we do this?
- Because we are now expressing in grams.
- 2 is the number of electrons that is getting exchanged over here and 9650 is zero is known to us.
- What is the value in ampere second?
- This value which you got from here is ampere seconds.

Example or application from the lecture

- Let us look at the solution.
- So we have to use this formula for capacity, WNF by M, because the weight is of lithium, which is used as the anodic material, is given to us.
- Let us see another type of question where we need to find out the energy density and the power density.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 10:00–14:59

60 into 60.

Key details

- Because straight away you can't write this and divide by 3600.
- It doesn't make sense.
- if you write this as ampere seconds and then you show it as ampere hours that would make more sense.
- Doesn't matter, we can always write it like this.
- divided by weight of the battery.
- If it was asked in a grams, then you to need to convert this into grams.
- But we neither we don't we know the time but we don't know the current.
- But is there any other way I can use this formula?

- Yes, we know that I into t is the capacity of the battery.
- And that is exactly what we calculated here.
- That is employed, and then we divide it by 20 weight of the battery, and we get 129.4 watt hours per kg is the energy density.
- Understand, we don't know the value of current, you need not know also, because once the capacity.
- You can straight away in this formula use capacity.
- will be energy density divided by time because it is current into voltage divided by.
- Current into voltage divided by.
- The weight of the battery.
- That is the formula for power density.
- Current into voltage divided by weight of the battery.
- Current into time into voltage by weight of the battery.
- Power density, all you have to do is if the energy density divide it by time and you get your formula I into v by w that's exactly what we have here.
- The time is known to us as 10 hours.
- This answer is also in hours.
- Instead we wanted it in hours because the time is in hours, they get the units get hour gets cancelled.
- here and therefore we'll end up with 129.4 divided by 10 and the answer would be this is also hours this is also what hour per kg so therefore you end up with watts per kg 12.9 watts per kg is your answer for power density.
- The efficiency of a fuel cell.
- We know that in an alkaline fuel cell.

Example or application from the lecture

- Volta is given to us as two volts here.
- And what is given to you E naught cell is given to you and delta H formation of water is equal to minus two eighty five point eight kilojoules per mole.
- Energy being given out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 15:00–19:59

with hydrogen and oxygen.

Key details

- H2O since we need only one mole, so this is how we write, and we know if you we know that if we write it as anode and cathode reaction, the number of electrons is going to be two electrons which will be exchanged in this reaction.
- Efficiency percentage is equal to delta G by delta H multiplied by 100.
- So when you substitute here and because we have to find out the percentage multiplied by 100.

- Minus N F E.
- We have to calculate the heat evolved during the reaction.
- Efficiency is again in the form of percentage.
- Efficiency percentage is delta G by delta H multiplied by 100.
- N again we know is going to be.
- Two for this particular reaction, what is the reaction happening here?
- Alkaline fuel cell H two plus half O2 is giving you.
- The delta H value is to be found out multiplied by 100.
- That's you will get it in joules, but then you want to determine in kilojoules.
- Both the reactions.
- When you say delta H combustion of hydrogen, when you say Delta H combustion.
- Or you can say delta H formation of one mole of water.
- In that case also the reaction remains the same, therefore the delta H will be same.
- So whether you say delta H formation of water or you say delta H combustion of hydrogen the value will remain the same and both of them here we see it is one mole therefore delta H is equal to minus 286 kilojoules per mole.

Example or application from the lecture

- Let us look at the solution.
- Now since here E naught cell is given to you and not delta G, we need to first calculate delta G.
- So this would give us the value of delta G and delta H is given to us minus 285.8 and because it's in kilojoules we will convert it into joules.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 20:00–24:59

You need to know the efficiency and you also need to know what is the E naught cell, or directly if the delta G also, you will be able to use this formula to calculate.

Key details

- So in the next class, we are going to start with the next unit that is about corrosion chemistry.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Numericals_Energy storage devices transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

C Revision and retrieval practice